



Inteligencia Artificial y Aprendizaje Automático

Sesión 1: Introducción a la
Inteligencia Artificial

Presentan:

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Fernando Ramírez González



HPC
DATA LAB

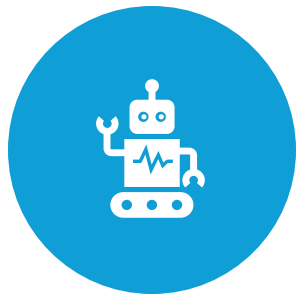
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Curso



2. ¿Qué es la
Inteligencia Artificial?



3. Aprendizaje
Automático (Machine
Learning)



4. Instalación y
Configuración de
Software



Presentación



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Actividad en Menti

<https://www.menti.com/alyxzfwt92fo>

1. Presentación del curso

Curso teórico-práctico

Revisión de
fundamentos
teóricos

Aplicaciones
prácticas
•Casos de
estudio reales

Programas
piloto en las
dependencias



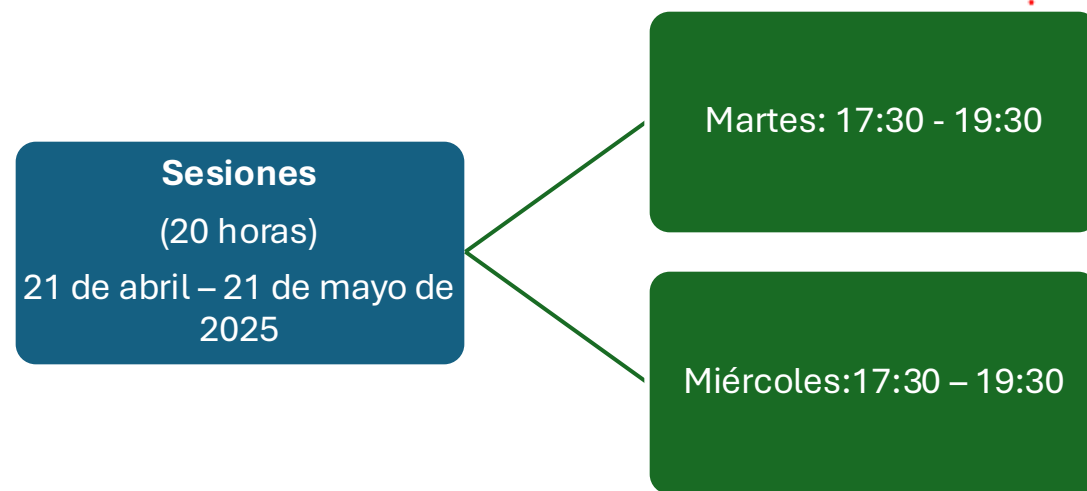


1. Presentación del curso

Objetivo General

Desarrollar una comprensión profunda y aplicada de los principios y técnicas fundamentales de la inteligencia artificial y el aprendizaje profundo, capacitando a los participantes para diseñar, implementar y evaluar soluciones avanzadas en diversos contextos y problemas reales.

1. Presentación del curso



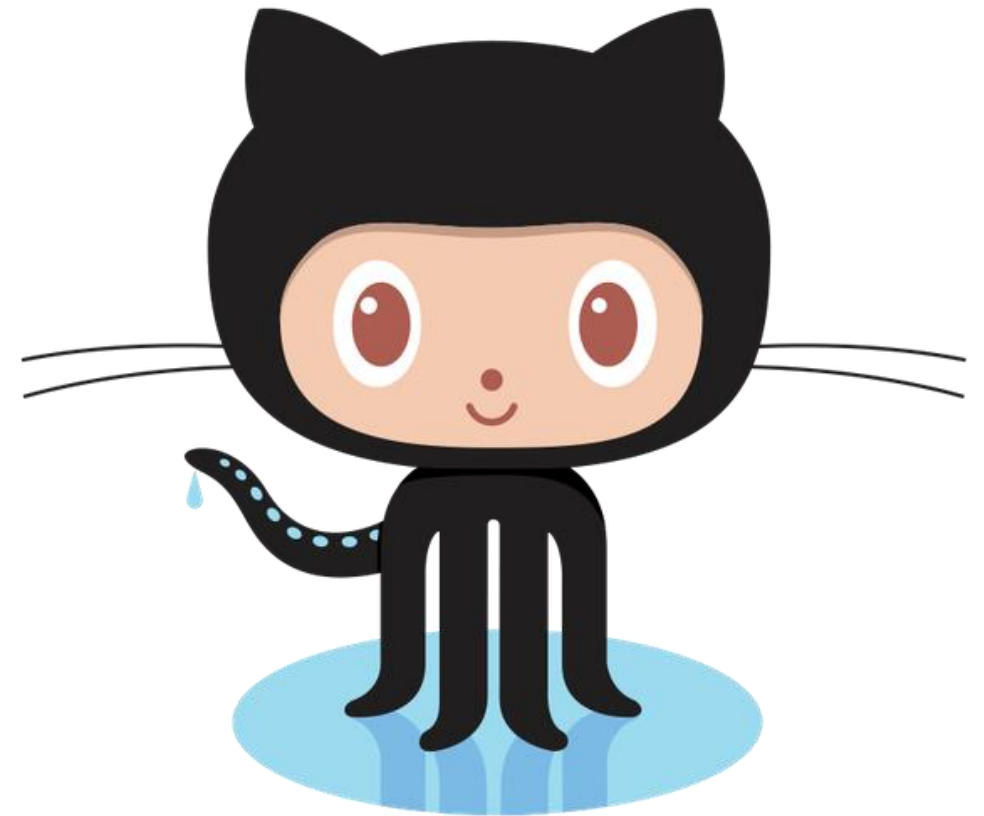
1. Presentación del curso

Plataformas

knnogllw



<https://classroom.google.com/c/NzcyOTk5MjA0MjM5?hl=es&cjc=knnogllw>

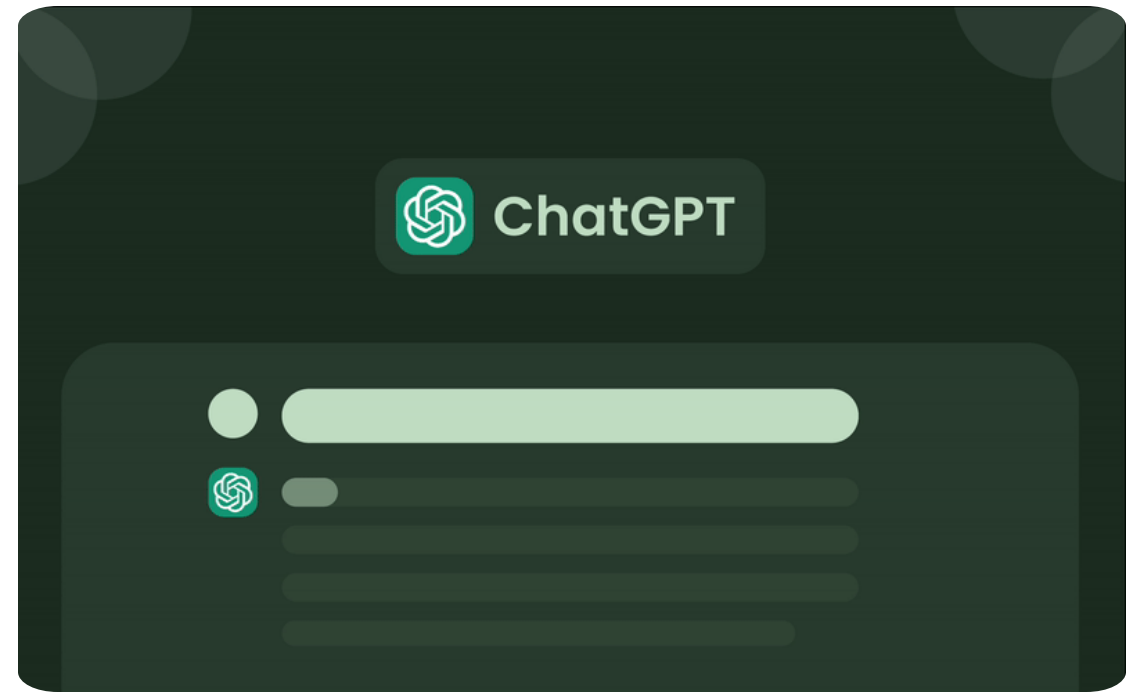


<https://github.com/ulises1229/IA-BLOQUE-2025>



Dudas, comentarios o preguntas

1. ¿Qué es la Inteligencia Artificial?



¿Qué es y cómo se define la inteligencia artificial (IA)?



Es un tópico con mucha relevancia actualmente.



Gran parte de la población mundial es usuaria de la IA.



La IA tiene distintas interpretaciones y significados para distintas personas.

Casi cualquier tipo de análisis de datos

Formas de vida artificiales





¿Qué si es IA y qué no es IA?

No existe una definición consensuada

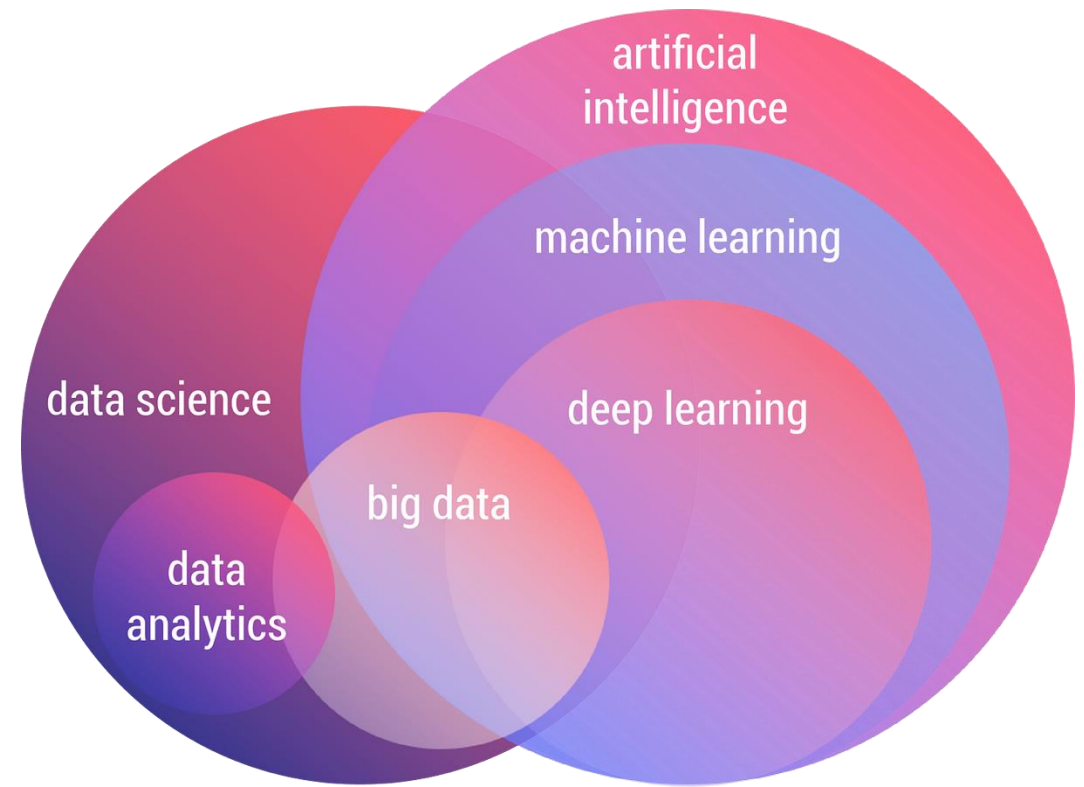
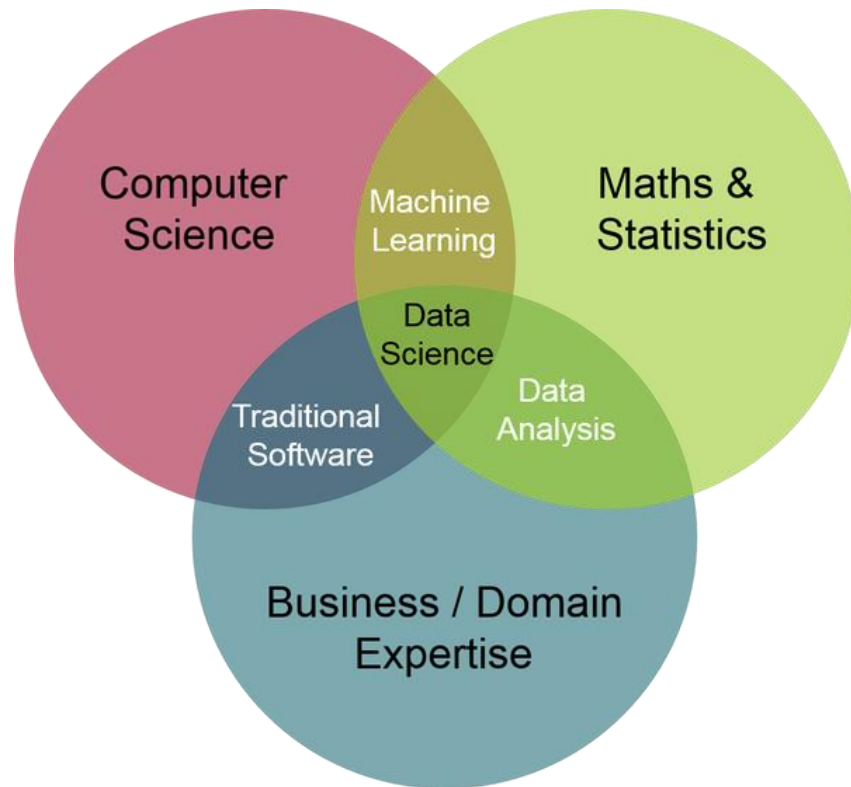
Existe un gran legado de la ciencia ficción

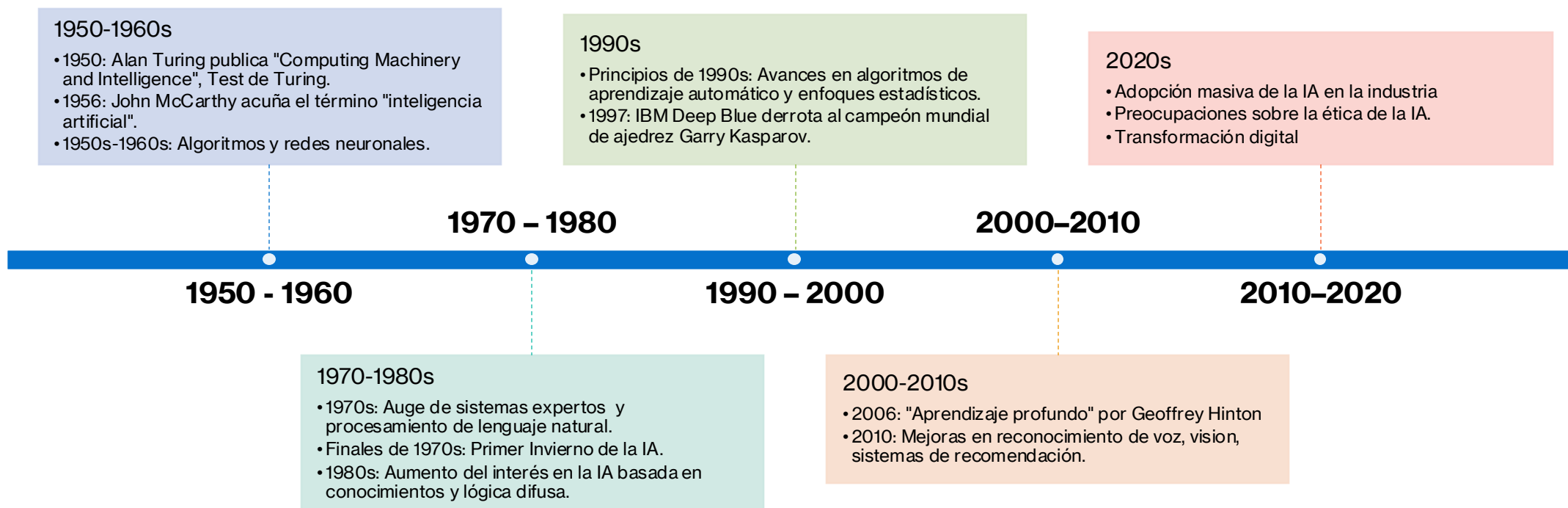
Lo que parece fácil, en realidad es muy complejo de imitar.

Lo que parece difícil, en realidad es muy fácil.



¿Qué es la Inteligencia Artificial?

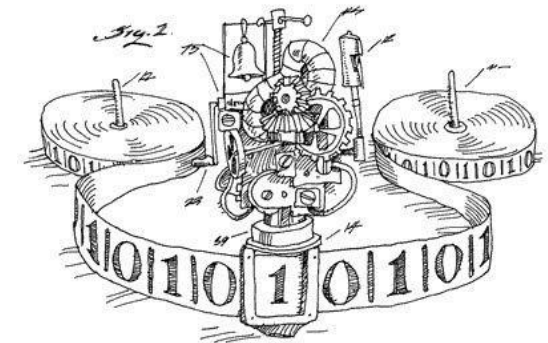
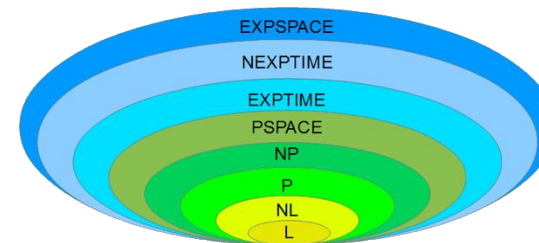
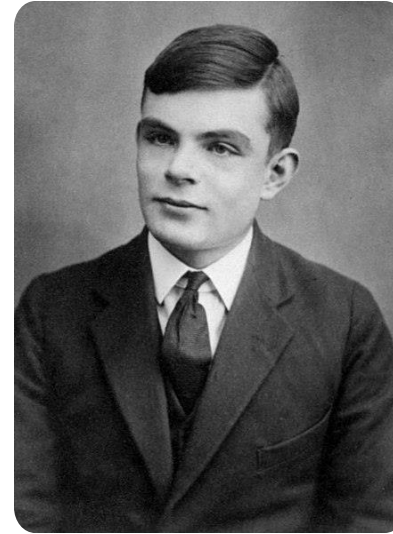


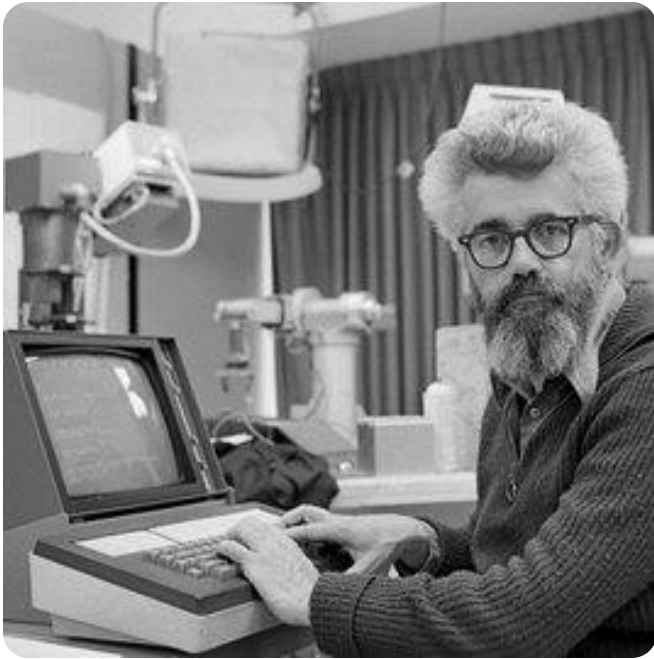


Historia de la IA

El test de Turing

- Alan Mathison Turing (1912 - 1954), matemático inglés considerado como el padre de la ciencia de la computación.
- Contribuciones:
 - Teoría de la computabilidad (Tesis Church-Turing)
 - Máquinas de Turing
 - Análisis Criptográficos (Enigma 1939)
 - Imitation game - Test de Turing (1950)
 - Eugene Goostman (13 años) - 10 de 30 personas.



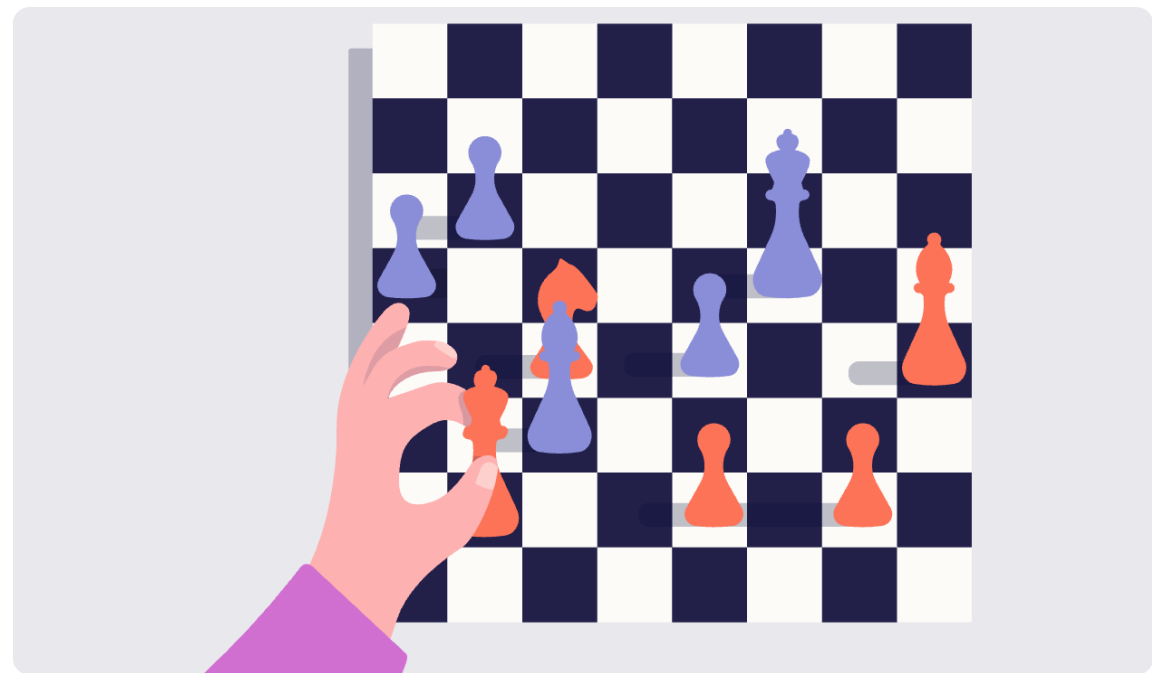
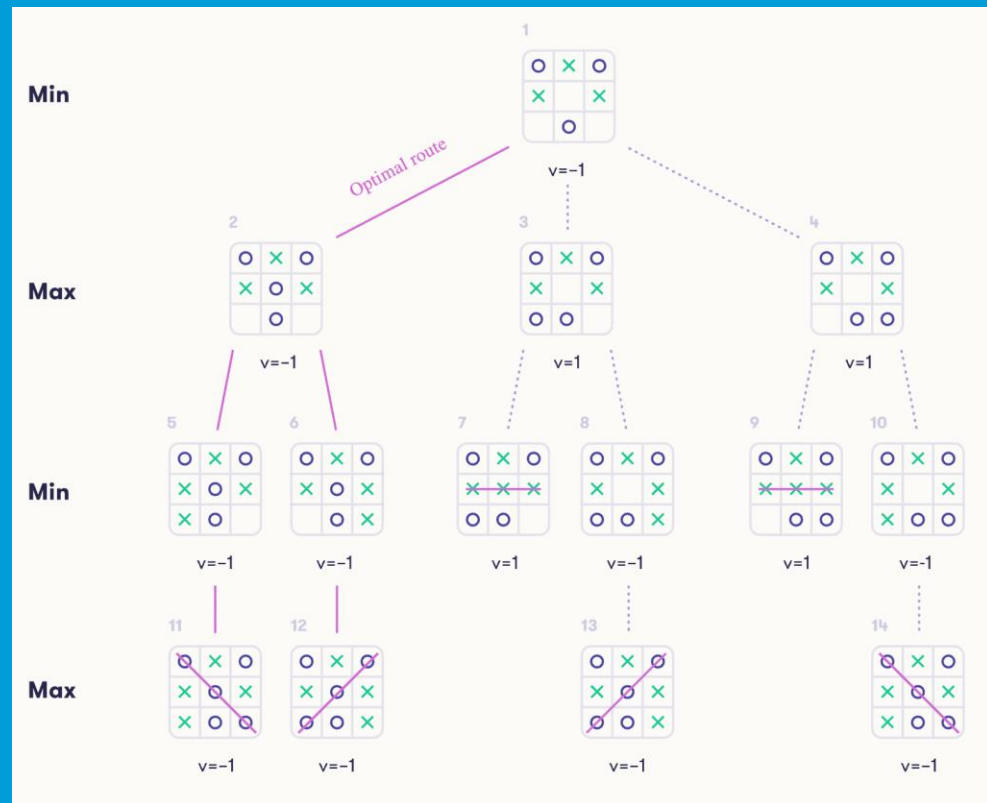


Resolución de problemas

- John McCarthy (1927-2011) considerado como el padre de la inteligencia artificial.

"Cualquier elemento de inteligencia puede dividirse en pequeños pasos para que cada uno de los pasos sea tan simple y "mecánico" que pueda escribirse como un programa de computadora".

- **Teoría de juegos (damas, ajedrez, etc.)**
- **Búsqueda y planificación**



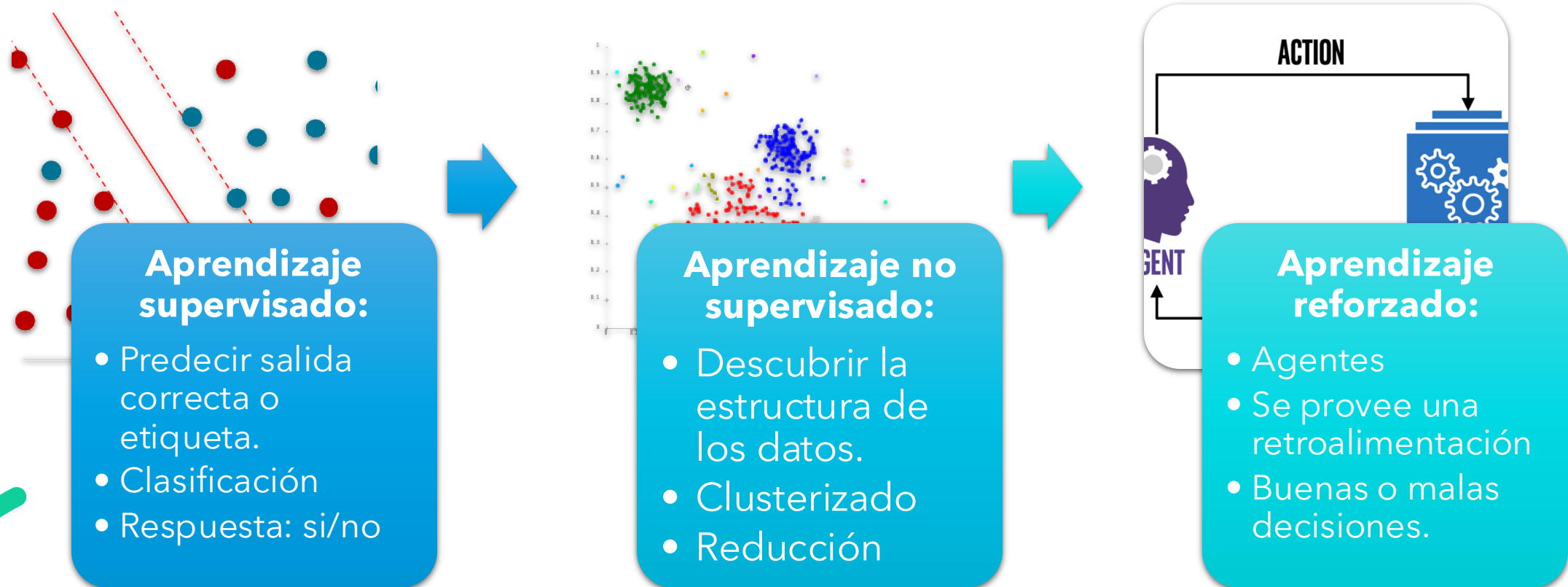
Incertidumbre a través de **probabilidad**

- Una de las razones por la que los algoritmos de IA actuales funcionan, es por su capacidad para lidiar con la incertidumbre. En comparación con los algoritmos de búsqueda que asumían escenarios estáticos (1960-1980).
- En la mayoría de las aplicaciones del mundo real, la única constante es la **incertidumbre** (probabilidad).



Aprendizaje automático (Machine Learning)

Extraer conocimiento de los datos



Aprendizaje Clásico



Regresión Lineal



Regresión Logística



Árboles de Decisión



Random Forest

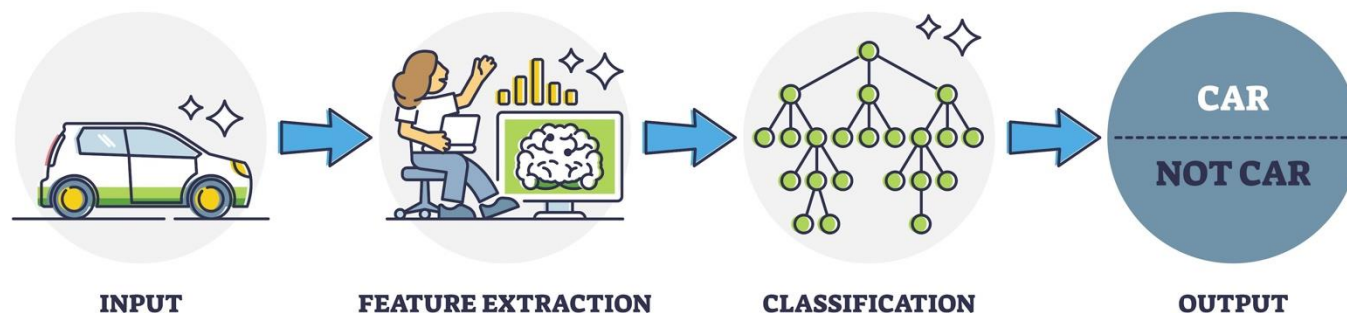


Support Vector Machines (SVM)

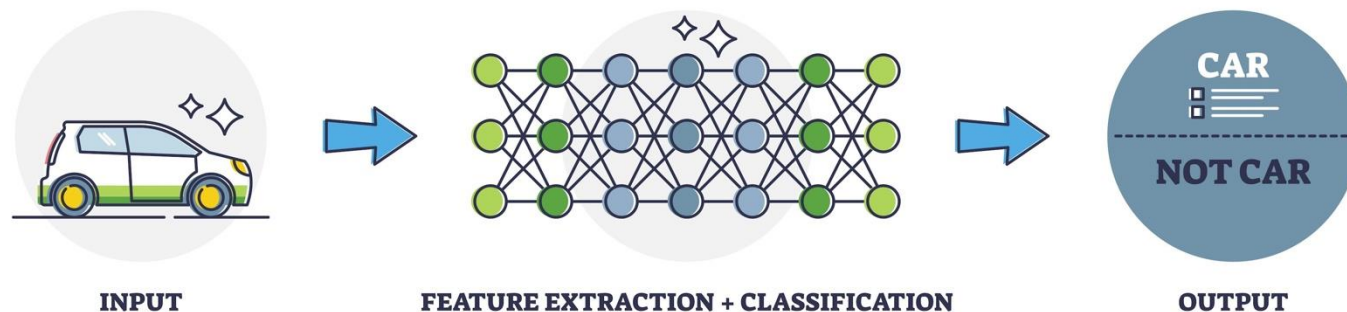


Naive Bayes

MACHINE LEARNING

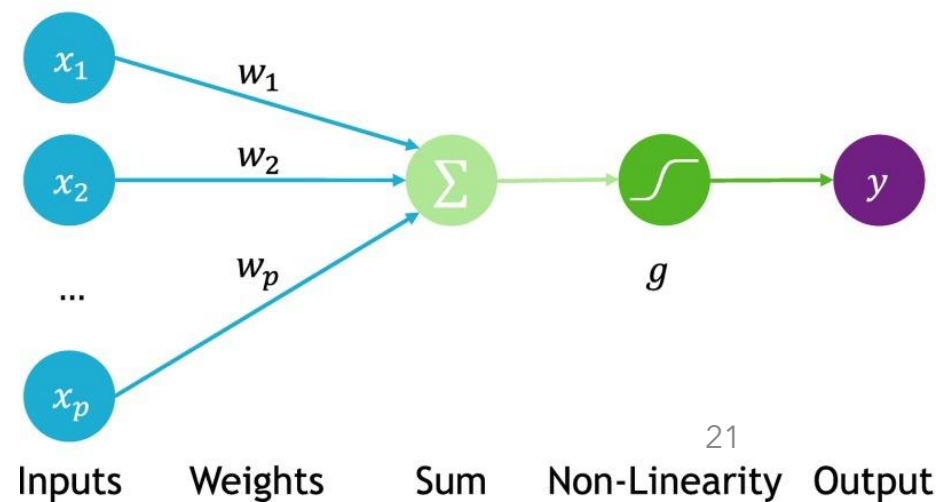
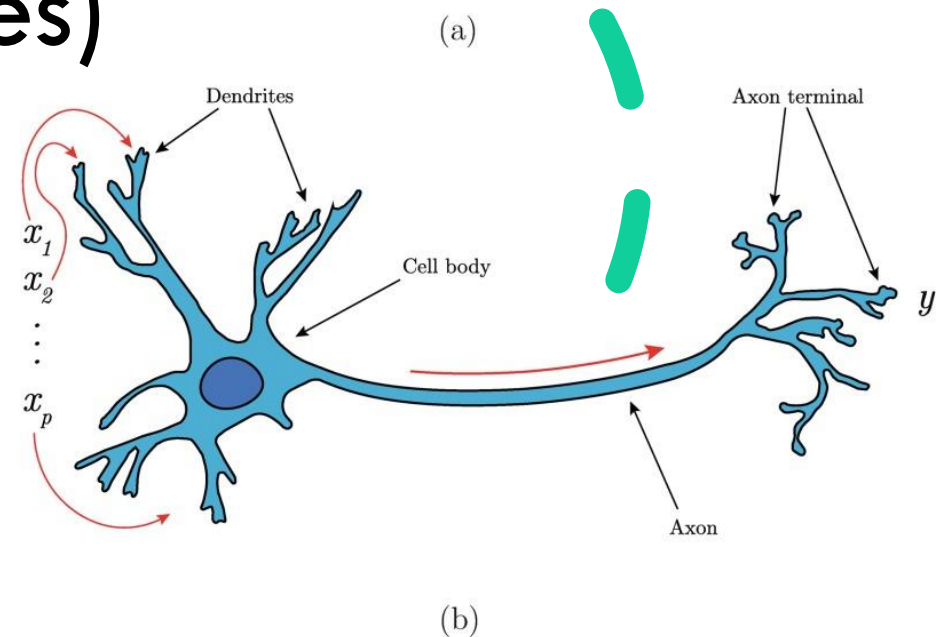
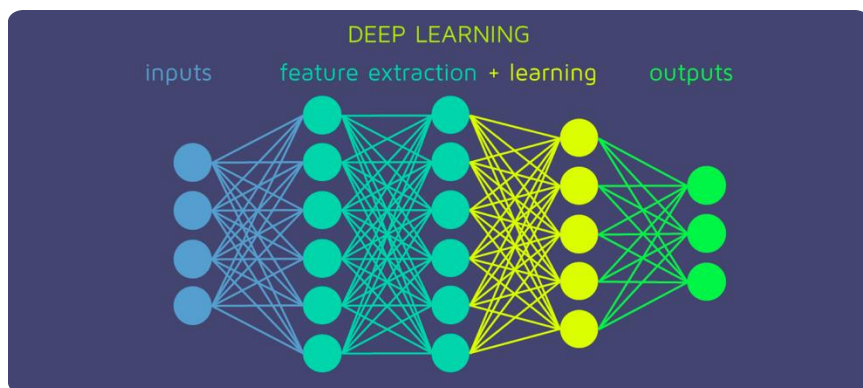


DEEP LEARNING



Algoritmos de Deep Learning (Redes Neuronales)

Deep Learning: se le conoce a ciertos tipos de técnicas de aprendizaje automático en las que varias **"capas"** de unidades de procesamiento simples están conectadas en una red para que la entrada al sistema pase a través de cada una de ellas.



Tipos de Redes Neuronales

A mostly complete chart of Neural Networks

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Backfed Input Cell

Input Cell

Noisy Input Cell

Hidden Cell

Probabilistic Hidden Cell

Spiking Hidden Cell

Output Cell

Match Input Output Cell

Recurrent Cell

Memory Cell

Different Memory Cell

Kernel

Convolution or Pool

Perceptron (P)



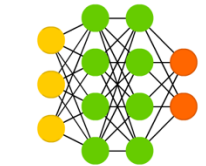
Feed Forward (FF)



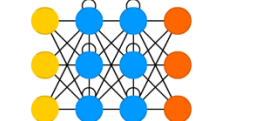
Radial Basis Network (RBF)



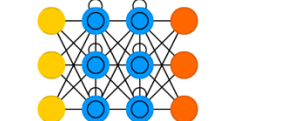
Deep Feed Forward (DFF)



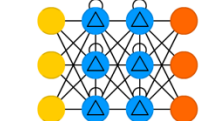
Recurrent Neural Network (RNN)



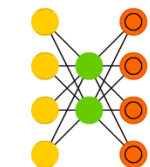
Long / Short Term Memory (LSTM)



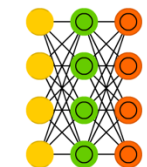
Gated Recurrent Unit (GRU)



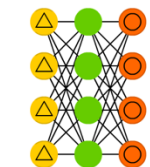
Auto Encoder (AE)



Variational AE (VAE)



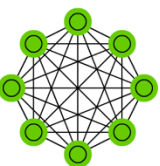
Denoising AE (DAE)



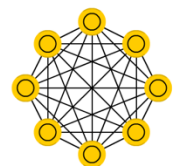
Sparse AE (SAE)



Markov Chain (MC)



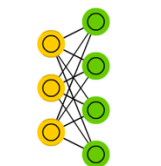
Hopfield Network (HN)



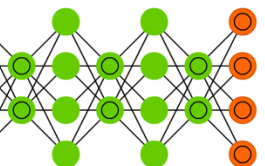
Boltzmann Machine (BM)



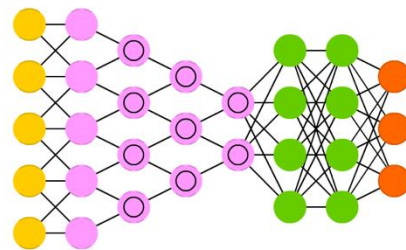
Restricted BM (RBM)



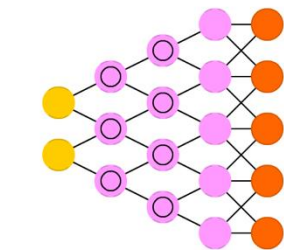
Deep Belief Network (DBN)



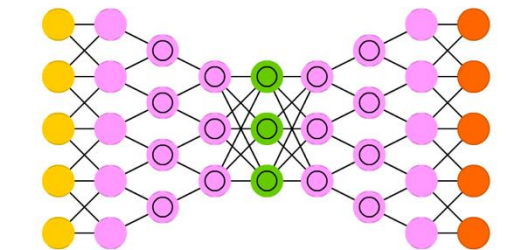
Deep Convolutional Network (DCN)



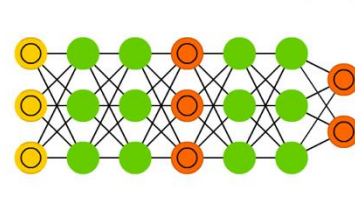
Deconvolutional Network (DN)



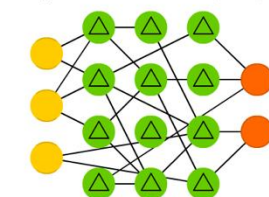
Deep Convolutional Inverse Graphics Network (DCIGN)



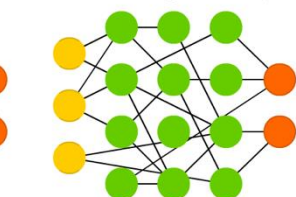
Generative Adversarial Network (GAN)



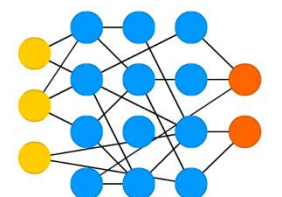
Liquid State Machine (LSM)



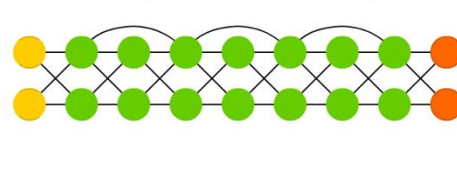
Extreme Learning Machine (ELM)



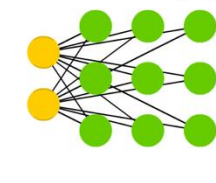
Echo State Network (ESN)



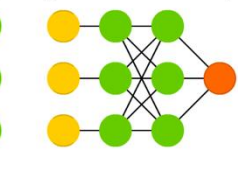
Deep Residual Network (DRN)



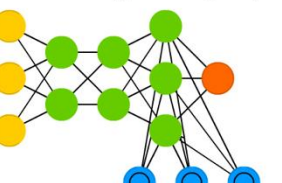
Kohonen Network (KN)



Support Vector Machine (SVM)

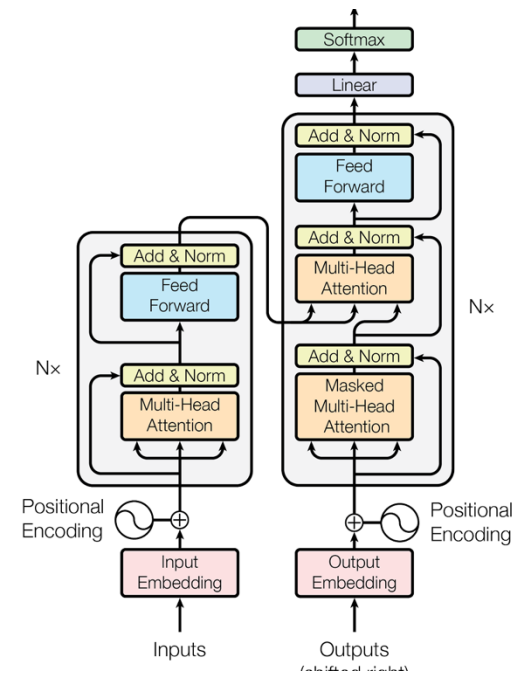


Neural Turing Machine (NTM)



Mecanismo de Atención

Arquitectura de Transformers



Attention Is All You Need

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Niki Parmar*
Google Research
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Jakob Uszkoreit*
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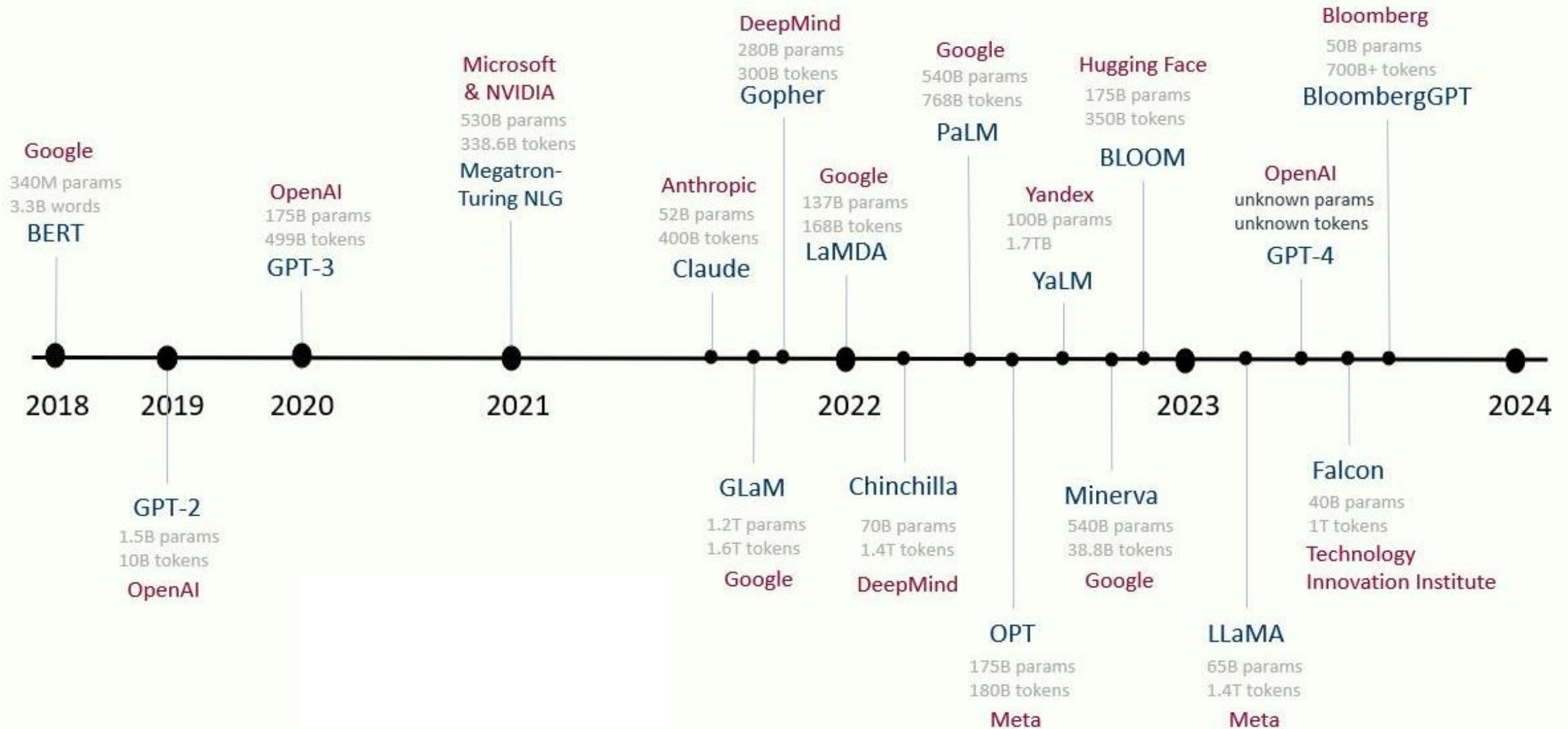
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aidan@cs.toronto.edu

Łukasz Kaiser*
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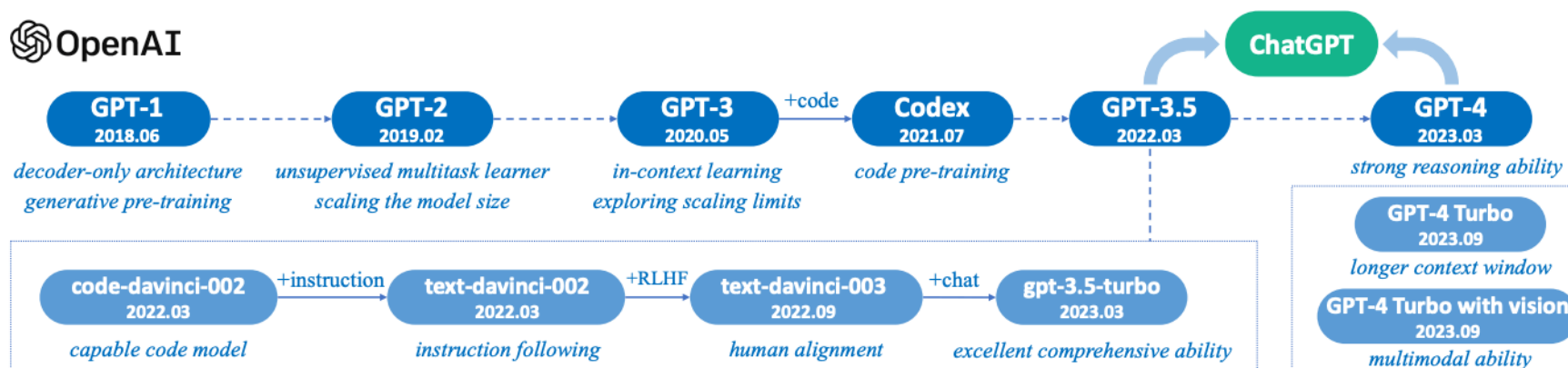
Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Selected LLMs

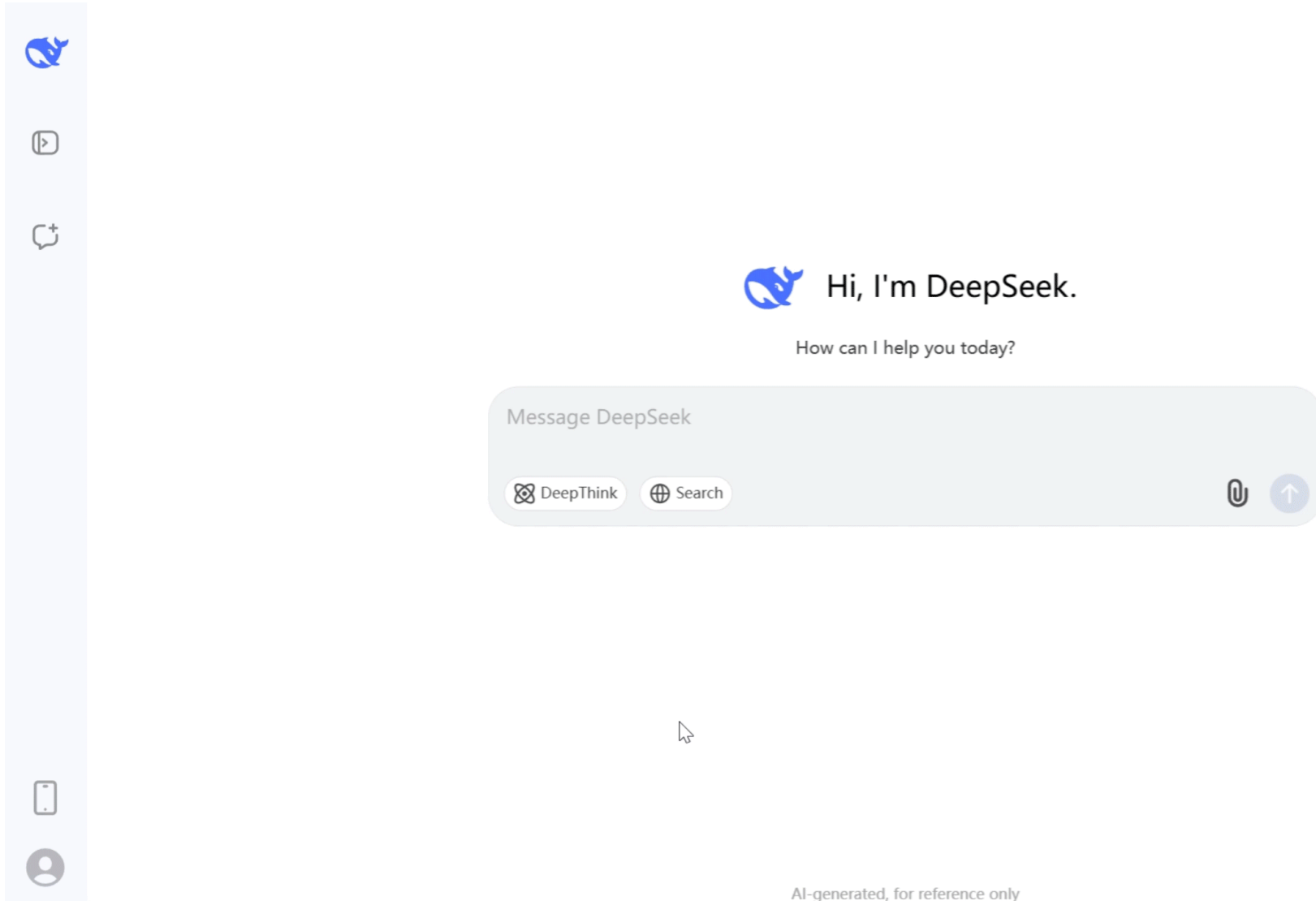


OpenAI

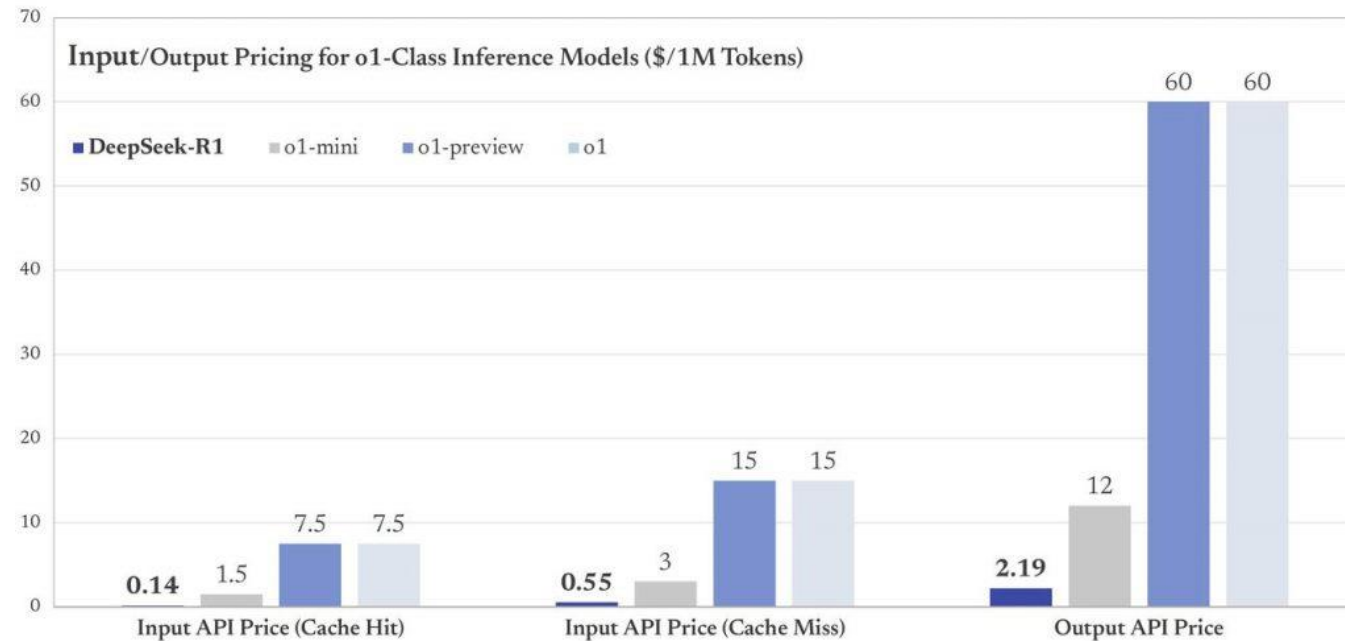
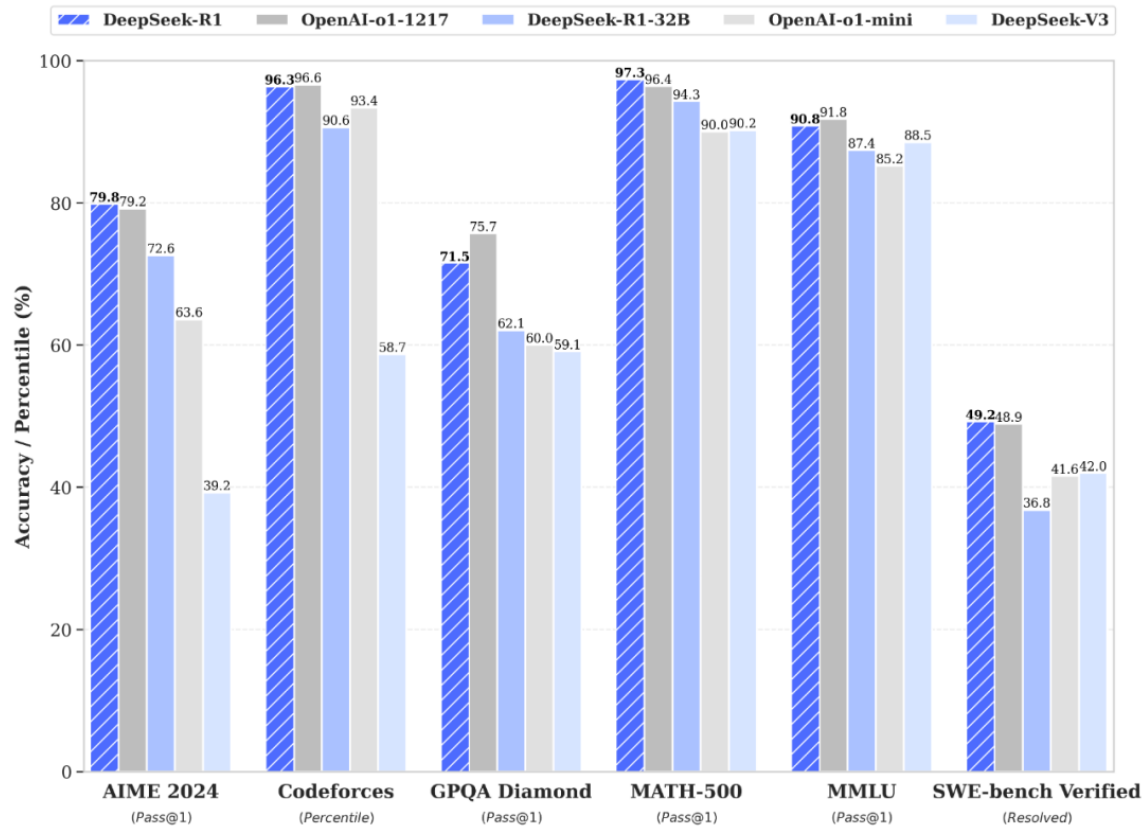
 OpenAI



DeepSeek V3 (2025)



DeepSeek Rendimiento y Costo



Avances Recientes



Figure

La startup de robótica Figure recauda 675 mdd de Microsoft, Nvidia y OpenAI

Figure, con sede en Sunnyvale, California, también dijo que firmó una colaboración con OpenAI para desarrollar inteligencia artificial generativa para sus robots humanoides.

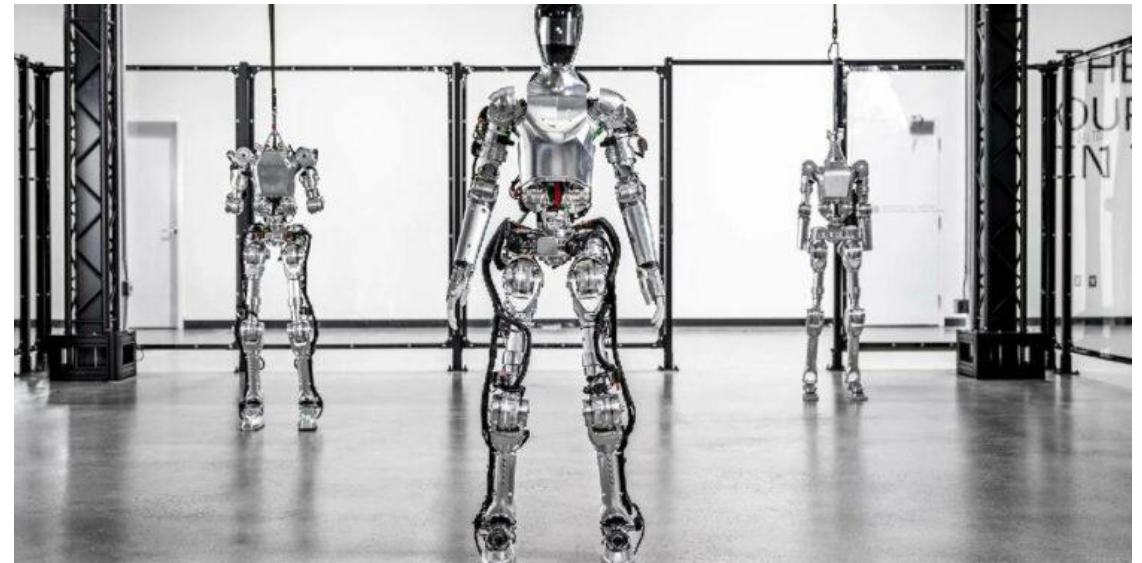


Figure Robot Humanoide



Figure Robot Humanoide



Figure Robot Humanoide



Open AI - Sora

Creating video from text

Sora is an AI model that can create realistic and imaginative scenes from text instructions.

[Read technical report](#)

All videos on this page were generated directly
by Sora without modification.

Open AI Sora

We're sharing our research progress early to get feedback from people outside of OpenAI and to give people a sense of what AI capabilities are on the horizon.

We will be taking several important safety steps before this research becomes available in any of our products.

Sora is a new AI model that can create realistic and imaginative scenes from text prompts.



Entrenamiento de modelos y robots

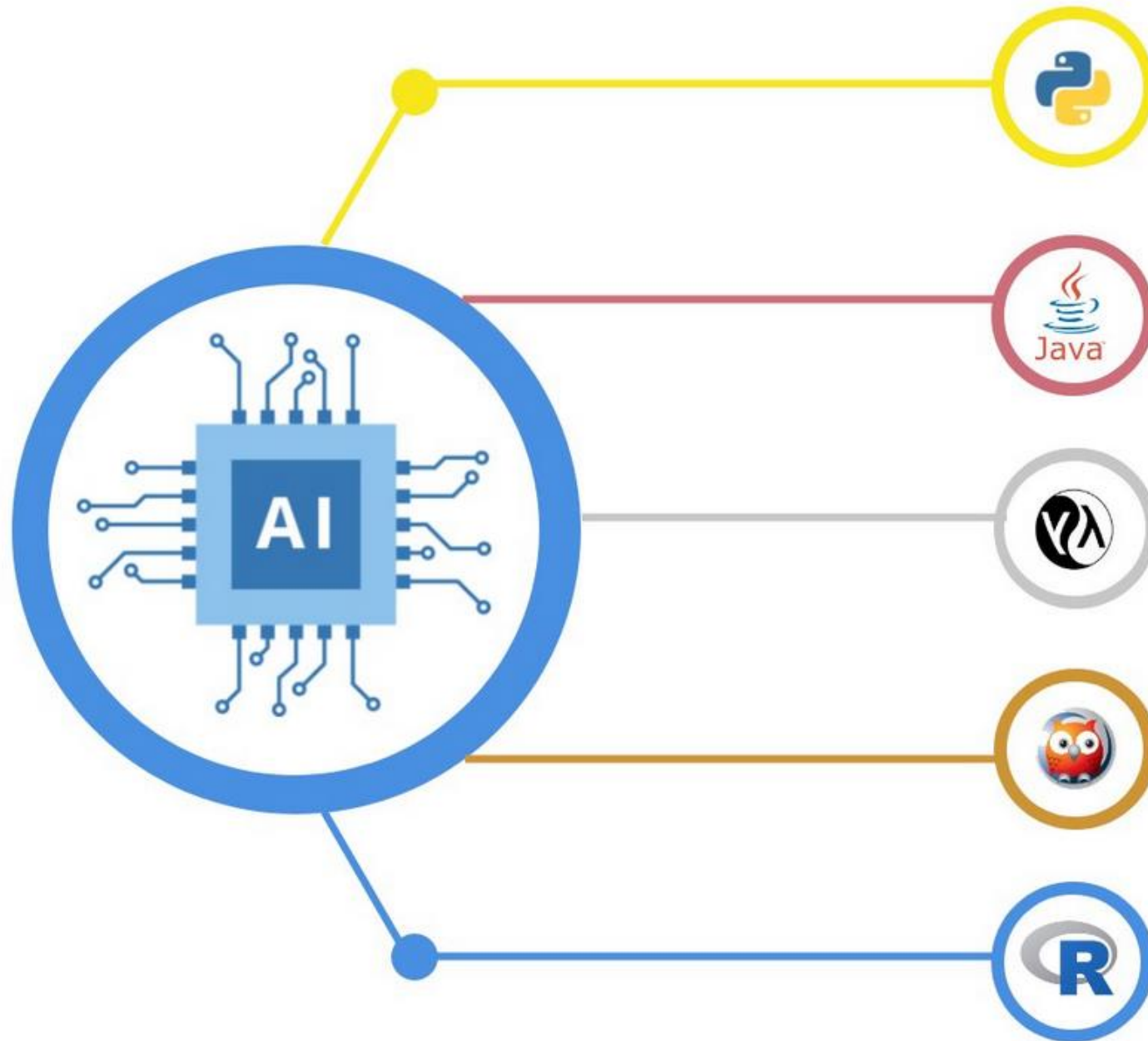




Demos y aplicaciones

IA generativa

Fusión de sensores



Python

Easy to learn, simple syntax and a lot of frameworks and libraries.



Java

Java is also be considered as a good choice For AI development



Lisp

One of the oldest and the most suited languages for the development in AI



Prolog

One of the logic programming language specifically designed for AI development.



R Programming

Comprehensive statistical language that encourages the development of new ideas.

kaggle™



kaggle

+ Create

- Home
- Competitions
- Datasets
- Models
- <> Code
- Discussions
- Learn
- More

✓ Your Work

viewed

- Cats and Dogs
- Facial Emotion Recog...
- SLIA - Lab 6 - Semant...
- Massachusetts Buildin...
- Models for Image Dat...

EDITED

- notebook990d875e85
- notebook83812a81c5

Search

Welcome, Ulises Olivares!

Jump back in, or start something new.

LOGIN STREAK: 1 (Your longest is 4 days)

TIER PROGRESS: 50% to Contributor

PUBLIC ACTIVITY: M T W T F S S

Datasets	Notebooks	Competitions	Discussions	Courses
1 total created	19 total created	2 total joined	0 total posted	0 total completed

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Welcome To Colab

File Edit View Insert Runtime Tools Help

Commands + Code + Text Copy to Drive

Table of contents

- Welcome to Colab!
- Getting started
- Data science
- Machine learning
- More Resources
- Featured examples
- + Section

Welcome to Colab!

Explore the Gemini API

The Gemini API gives you access to Gemini models created by Google DeepMind. Gemini models are built from the ground up to be multimodal, so you can reason seamlessly across text, images, code, and audio.

How to get started?

- Go to [Google AI Studio](#) and log in with your Google account.
- Create an API key.
- Use a quickstart for [Python](#), or call the REST API using [curl](#).

Discover Gemini's advanced capabilities

- Play with Gemini [multimodal outputs](#), mixing text and images in an iterative way.
- Discover the [multimodal Live API](#) (demo [here](#)).
- Learn how to [analyze images and detect items in your pictures](#) using Gemini (bonus, there's a [3D version](#) as well!).
- Unlock the power of [Gemini thinking model](#), capable of solving complex task with its inner thoughts.

Explore complex use cases

- Use [Gemini grounding capabilities](#) to create a report on a company based on what the model can find on internet.
- Extract [invoices and form data from PDF](#) in a structured way.
- Create [illustrations based on a whole book](#) using Gemini large context window and Imagen.

To learn more, check out the [Gemini cookbook](#) or visit the [Gemini API documentation](#).

Colab now has AI features powered by [Gemini](#). The video below provides information on how to use these features, whether you're new to Python, or a seasoned veteran.



3. Instalación y configuración de software





ANACONDA®

Instalación de Ambiente de Desarrollo

Python y Jupyter y
librerías

<https://www.anaconda.com/>

Anaconda

- La manera más sencilla de instalar **Python**, **Jupyter Notebook** y librerías esenciales como **NumPy** y **pandas** es a través de la distribución **Anaconda**.
- Anaconda agrupa convenientemente **Python**, **Jupyter Notebook** y otros paquetes comúnmente utilizados para la computación científica y la ciencia de datos.



ANACONDA®

Environments

Learning

Community

Anaconda Toolbox
Supercharged local notebooks. Click the Toolbox tile to install.

[Read the Docs](#)

[Documentation](#)

[Anaconda Blog](#)



PyCharm Professional

A Full-fledged IDE by JetBrains for both Scientific and Web Python development. Supports HTML, JS, and SQL.

[Install](#)



Anaconda Toolbox

0.4.0

Anaconda Assistant

JupyterLab supercharged with a suite of Anaconda extensions, starting with the Anaconda Assistant AI chatbot.

[Install](#)



Anaconda Cloud Notebooks

Cloud-hosted notebook service from Anaconda. Launch a preconfigured environment with hundreds of packages and store project files with persistent cloud storage.

[Launch](#)



Notebook

[7.0.6](#)

Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.

[Launch](#)



Qt Console

[5.4.2](#)

PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more.

[Launch](#)



Spyder

[5.4.3](#)

Scientific PYTHON Development EnviRnment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features

[Launch](#)



VS Code

1.87.2

Streamlined code editor with support for development operations like debugging, task running and version control.

[Launch](#)



Anaconda on AWS Graviton

Running your Anaconda workloads on AWS Graviton-based processors could provide up to 40% better price performance

[Launch](#)



Oracle Data Science Service

OCI Data Science offers a machine learning platform to build, train, manage, and deploy your machine learning models on the cloud with your favorite open-source tools

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Glueviz

1.2.4

Multidimensional data visualization across files. Explore relationships within and among related datasets.

[Install](#)



JupyterLab

4.0.11

An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.

[Install](#)



Orange 3

3.34.0

Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.

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